

SOLAR ENERGY FORECASTING

Case Study Report

**SDG 7 – Affordable & Clean Energy | SDG 13 – Climate
Action**

Solar Power Generation Dataset – Plants 1 and 2 with
Weather Sensor Data

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Abstract

Solar photovoltaic generation is variable because the available solar resource changes with time and weather conditions. This case study analyses generation and weather-sensor data from two solar plants and develops a short-term forecasting approach using time-series features, recent generation history and weather variables. The analysis covers data from 15 May 2020 to 17 June 2020. The study examines daily and hourly generation patterns, the relationship between irradiation and power output, temperature effects, and correlations among important variables. A Random Forest regression model is evaluated using a chronological 80:20 train-test split. On the held-out test period, the model obtained MAE = 467.41, RMSE = 1208.99, and $R^2 = 0.9693$. The results demonstrate that combining recent generation history with weather and time-based information can provide a useful data-driven basis for short-term solar power forecasting.

The study performs exploratory data analysis to identify daily and hourly solar-generation trends and to compare the performance of the two plants. The influence of weather conditions on solar generation is examined using correlation analysis and visualizations such as scatter plots and correlation heatmaps. The analysis shows a strong positive relationship between solar irradiation and AC power, demonstrating the importance of available sunlight in determining photovoltaic output. Temperature-related variables are also examined to understand their association with power generation.

1. Problem Statement

Solar power generation changes throughout the day and across weather conditions. Unlike conventional generation sources, photovoltaic output cannot be treated as constant because it depends strongly on solar irradiation and the operating environment of the panels.

Accurate forecasting can support planning, scheduling, monitoring and better integration of renewable electricity. Therefore, the case study focuses on understanding the observed generation patterns, identifying weather-related influences and building a regression model for short-term AC-power forecasting.

Objectives

- Understand the structure and quality of the solar generation and weather datasets.
- Identify daily, hourly and monthly generation patterns for the two plants.
- Study the relationship between irradiation, temperature and AC power.
- Create time-based and lag-based features suitable for forecasting.
- Train and evaluate a Random Forest regression model using a chronological test split.
- Interpret the results in the context of renewable-energy planning and monitoring.

2. Literature Survey

Inman, Pedro and Coimbra (2013)

Solar forecasting methods for renewable energy integration. The review covers solar forecasting across different temporal scales and emphasizes the need for accurate forecasts when variable renewable generation is integrated into power systems. It also discusses the variability of the solar resource caused by atmospheric and weather-related factors.

Progress in Energy and Combustion Science, 39(6), 535–576. DOI: 10.1016/j.pecs.2013.06.002

Antonanzas et al. (2016)

Review of photovoltaic power forecasting. The study surveys forecasting horizons, inputs and methods for PV power prediction and explains why forecasting is important for grid operation, scheduling and economic dispatch.

Solar Energy, 136, 78–111. DOI: 10.1016/j.solener.2016.06.069

Voyant et al. (2017)

Machine learning methods for solar radiation forecasting: A review. The paper surveys machine-learning approaches for solar radiation forecasting and discusses how model performance depends on data, forecasting horizon and evaluation methodology.

Renewable Energy, 105, 569–582. DOI: 10.1016/j.renene.2016.12.095

Research Gap Addressed

The reviewed studies establish the importance of solar forecasting and the use of statistical, physical and machine-learning approaches. This case study applies a complete data-driven workflow to plant-level generation and weather measurements, including preprocessing, exploratory analysis, feature engineering, chronological validation and quantitative evaluation.

3. Implementation

The implementation follows a complete data-analysis workflow: loading the four source files, checking and preparing the data, aggregating plant generation, matching weather observations, performing exploratory analysis, engineering time-series features and evaluating a forecasting model.

3.1 Dataset Description

File	Main information
Plant_1_Generation_Data.csv	Timestamped DC power, AC power, daily yield and total yield for Plant 1.
Plant_1_Weather_Sensor_Data.csv	Ambient temperature, module temperature and irradiation for Plant 1.
Plant_2_Generation_Data.csv	Timestamped DC power, AC power, daily yield and total yield for Plant 2.
Plant_2_Weather_Sensor_Data.csv	Ambient temperature, module temperature and irradiation for Plant 2.

The final analytical variables include DATE_TIME, DC_POWER, AC_POWER, DAILY_YIELD, TOTAL_YIELD, AMBIENT_TEMPERATURE, MODULE_TEMPERATURE, IRRADIATION and PLANT.

3.2 Data Preprocessing

1. Converted DATE_TIME into a datetime format.
2. Removed duplicate observations before analysis.
3. Aggregated inverter-level AC and DC power by timestamp and plant.
4. Aggregated weather observations by timestamp and plant.
5. Merged generation and weather data using DATE_TIME and PLANT_ID.
6. Created hour, minute, day-of-week, month and day-of-year features.

7. Created 15-minute, one-hour and one-day lag features using previous AC-power observations.
8. Used a chronological 80:20 train-test split so that future observations were not mixed into the training data.

3.3 Exploratory Data Analysis

The analysis examined daily and hourly generation trends, plant comparison, irradiation versus AC power, temperature relationships and correlation patterns.

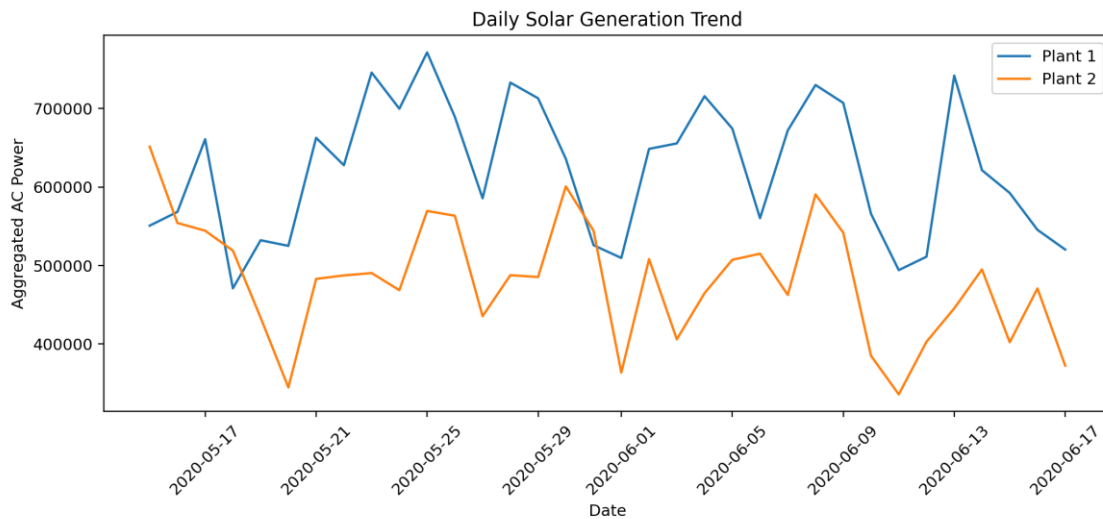


Figure 1. Daily solar generation trend.

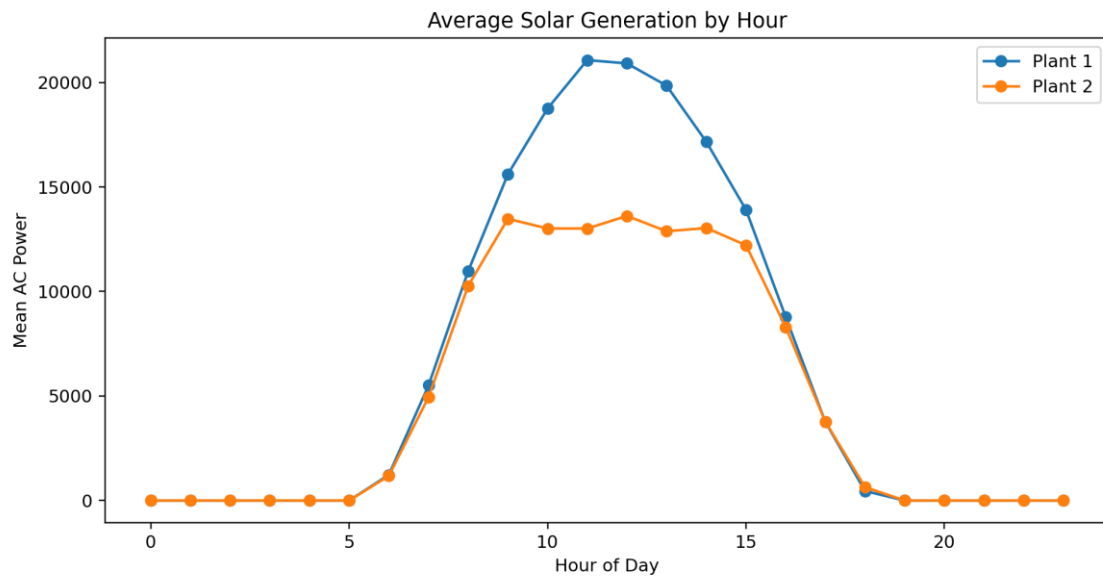


Figure 2. Average solar generation by hour.

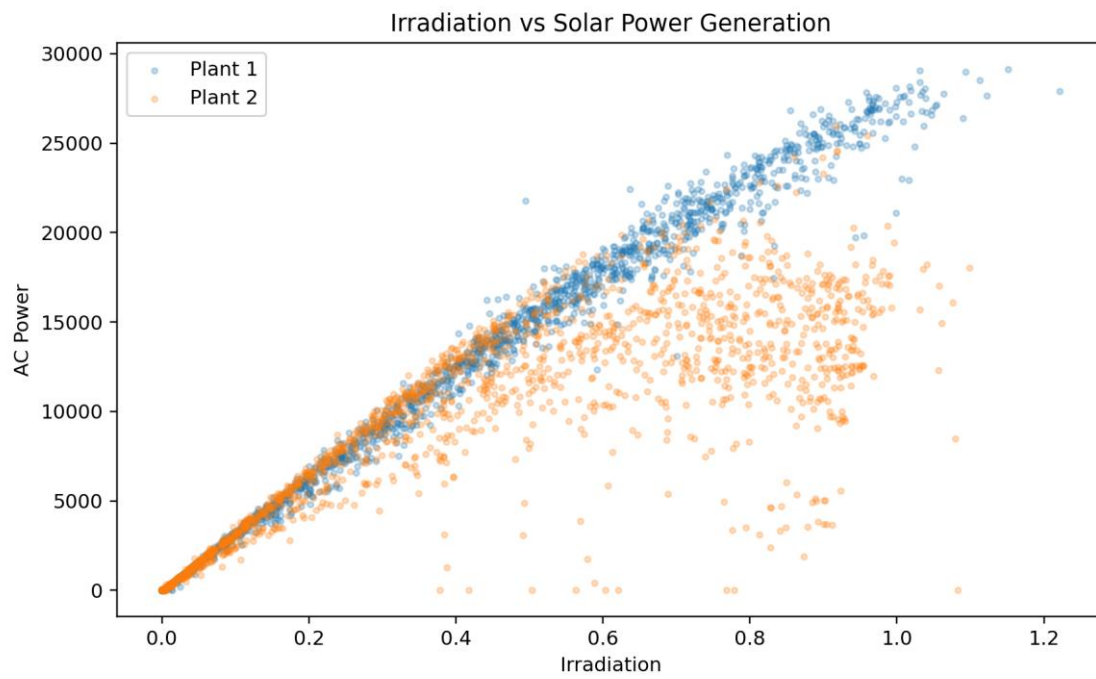


Figure 3. Irradiation versus AC power.

3. Implementation – Forecasting Model

3.4 Feature Engineering

The forecasting dataset combines weather variables with temporal and historical-generation information. Calendar features capture regular daily and seasonal patterns, while lagged AC power provides recent generation history. This makes the experiment suitable for short-term forecasting rather than long-range planning without recent observations.

3.5 Forecasting Model

A Random Forest Regressor was selected for the forecasting experiment. The model uses ambient temperature, module temperature, irradiation, calendar variables, plant identity and lagged AC power as predictors. The target variable is AC_POWER. Performance was evaluated with Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and coefficient of determination (R^2).

3.6 Evaluation Strategy

- Training data: first 80% of the chronological observations.
- Testing data: final 20% of the chronological observations.
- MAE: average absolute prediction error.
- RMSE: square-rooted mean squared error, giving more influence to large errors.
- R^2 : proportion of target variation explained by the model under the experimental setup.

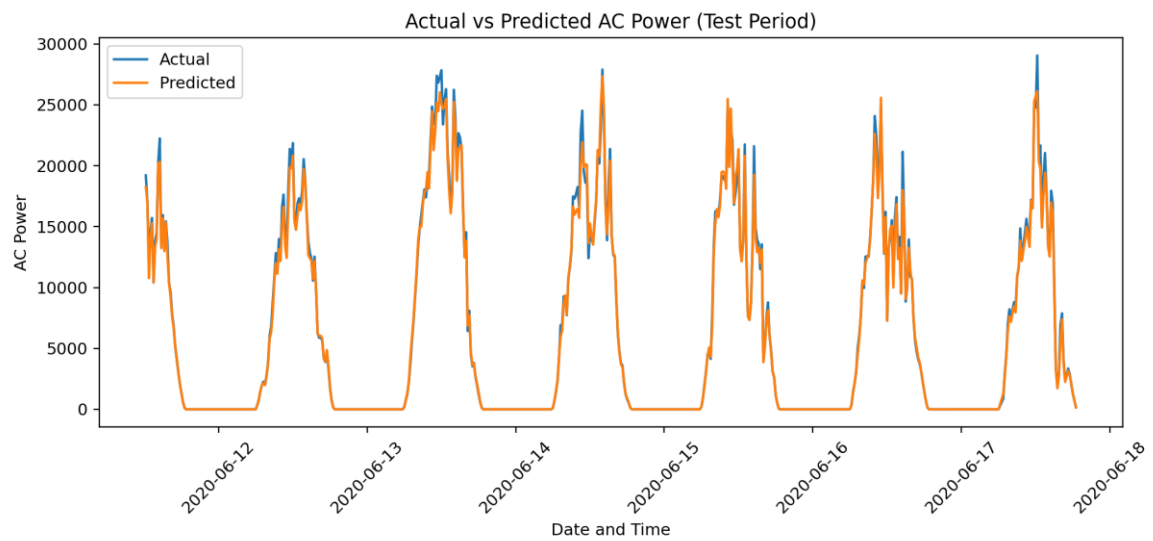


Figure 4. Actual versus predicted AC power on the held-out test period.

4. Results

4.1 Summary of Results

Measure	Result
Analytical dataset	6,416 timestamp-level observations after plant-level aggregation and weather matching.
Study period	15 May 2020 to 17 June 2020.
Plant 1 aggregated AC-power total	21,167,150 units in the analytical dataset.
Plant 2 aggregated AC-power total	16,334,026 units in the analytical dataset.
AC power vs irradiation	Strong positive association, $r = 0.938$.
AC power vs module temperature	Strong positive association, $r = 0.903$.
AC power vs ambient temperature	Positive association, $r = 0.593$.
Forecasting MAE	467.41 AC-power units.
Forecasting RMSE	1208.99 AC-power units.
Forecasting R^2	0.9693 on the held-out chronological test period.

4.2 Weather Impact

The correlation analysis shows a strong positive association between AC power and irradiation ($r = 0.938$) and between AC power and module temperature ($r = 0.903$). Ambient temperature also shows a positive association with AC power ($r = 0.593$). These values are descriptive correlations and should not be interpreted as proof of causation.

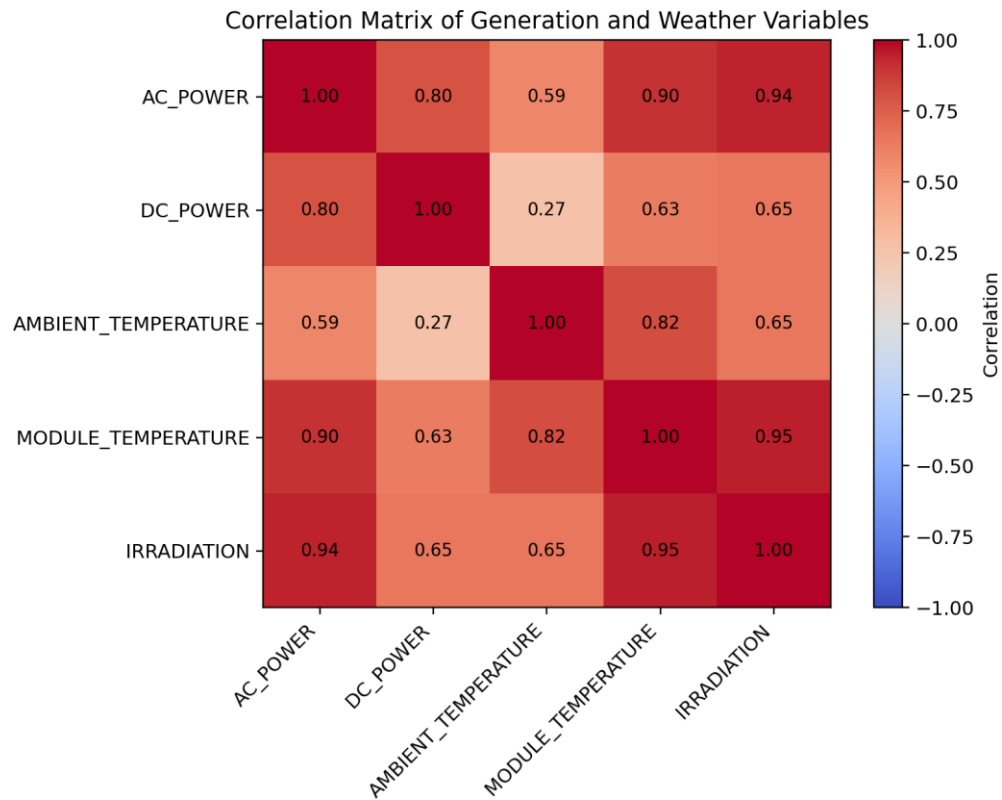


Figure 5. Correlation between weather variables and solar generation.

4.3 Forecasting Performance

The Random Forest model achieved $R^2 = 0.9693$ on the held-out chronological test period, with MAE = 467.41 and RMSE = 1208.99. The result indicates a strong fit for this experimental setup. Because lagged observed AC power is included as an input, the evaluation represents short-term forecasting where recent plant output is available.

5. Conclusion

This case study analysed solar power generation and weather-sensor observations from two photovoltaic plants. The workflow combined data cleaning, timestamp aggregation, weather matching, exploratory analysis and time-series feature engineering. The analysis shows clear daily and hourly generation patterns and a strong descriptive relationship between solar irradiation and AC power.

A Random Forest regression model using weather variables, time features and lagged generation produced MAE = 467.41, RMSE = 1208.99 and $R^2 = 0.9693$ on the chronological test period. The experiment indicates that historical generation and weather measurements can be combined to model short-term photovoltaic output. Such forecasting can support renewable-energy planning, monitoring and grid-integration activities.

Future work can extend the study with additional meteorological variables, longer forecasting horizons, alternative models such as Gradient Boosting or recurrent neural networks, uncertainty intervals and comparison against simple forecasting baselines.

References

9. Inman, R. H., Pedro, H. T. C., & Coimbra, C. F. M. (2013). Solar forecasting methods for renewable energy integration. *Progress in Energy and Combustion Science*, 39(6), 535–576. DOI: 10.1016/j.pecs.2013.06.002
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Dataset Files Used

- Plant_1_Generation_Data.csv
- Plant_1_Weather_Sensor_Data.csv
- Plant_2_Generation_Data.csv
- Plant_2_Weather_Sensor_Data.csv